GALGOTIAS UNIVERSITY

SCHOOL OF COMPUTING SCIENCE & ENGINEERING

PROJECT APPROVAL FORM AND ABSTRACT

Odd 2025-2026

B. Tech Project Details

Project ID: BT0644

Title	**Unified Health ID: Overcoming Fragmented records & Low ABHA Adoption**		
Project Type	☐ Community based design problem (Interdisciplinary) ☐ App Development / Utility ☐ IOT/Hardware based ☐ AI/ML/Data Science ☑ Healthcare Projects	**Project Outcome**	☑ Project and Research Paper ☐ Project and Patent ☐ Project and Book Chapter
		SDG Goal Number	3
Publication Target	☑ SCOPUS Journal ☐ SCOPUS Conference ☐ SCOPUS Book Chapter ☐ SCI Journal	**Guide Name: Dr.Firoj Ahamad**	

Student Details:

S. No	Name	Enrollment Number	Admission Number	Program / Branch	Sem
1	Gauri Pachauri	22131011550	22SCSE1011577	B.Tech CSE	VII
2	Harshit Rathore	22131011561	22SCSE1011589	B.Tech CSE	VII

Sustainable Development Goals (SDG Goals), Targets and Indicators

Goal 1: End poverty in all its forms everywhere
Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture
Goal 3: Ensure healthy lives and promote well-being for all at all agessaf
Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
Goal 5: Achieve gender equality and empower all women and girls
Goal 6: Ensure availability and sustainable management of water and sanitation for all
Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all
Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
Goal 10: Reduce inequality within and among countries
Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable
Goal 12: Ensure sustainable consumption and production patterns
Goal 13: Take urgent action to combat climate change and its impacts
Goal 14: Conserve and sustainably use the oceans, seas and marine resources for sustainable development
Goal 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
Goal 17: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

Scanned with OKEN Scanner

Unified Health ID: Overcoming Fragmented Records and Low ABHA Adoption

Area/Domain of Project: Web And Mobile App Development

ABSTRACT: India is witnessing a steady rise in chronic illnesses such as diabetes, hypertension, heart disease, and cancer, which together account for a large share of nationwide deaths. Although initiatives like the Ayushman Bharat Digital Mission (ABDM) and ABHA health ID have been launched to strengthen the healthcare system, several barriers limit their effectiveness. Key issues include fragmented health records, low digital adoption in rural regions, and the lack of integrated approaches that support long-term recovery.

These shortcomings create multiple challenges. Patients are often subjected to repeat diagnostic tests, delayed care, higher treatment costs, and limited participation in managing their own health. Doctors lack continuous access to patient histories, while researchers struggle to obtain structured and anonymized data for meaningful studies.

To overcome these gaps, this project proposes a **patient-centric digital health identification system**. The system would consolidate lifelong medical records from birth, support secure and consent-based data sharing, and include personalized, evidence-driven lifestyle and natural care suggestions in addition to conventional treatment.

A **web-based patient portal** and **clinician dashboard** will be developed to demonstrate functionality and usability. Anticipated outcomes include improved continuity of care, reduced duplication in testing and spending, empowered patients through personalized insights, and the availability of reliable datasets for medical research and policymaking. By bridging these gaps, the project aims to strengthen India's digital health ecosystem and support sustainable development goals.

Problem Statement: India's HealthCare System faces fragmented records, poor ABHA adoption, and weak chronic disease management, a patient-centric digital health ID with lifelong, consent-driven records can improve care continuity and outcomes.

30-09-25

Signature of Students **Signature of Guide**

Student 1 Name: Gauri Pachauri

Gauri Pachauri 30/9/25

Student 2 Name: Harshit Rathore

Harshit Rathore 30-09-25

Scanned with OKEN Scanner

A Project Report

On

UHID: Unified Health ID

Submitted in partial fulfillment of the

requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

GALGOTIAS UNIVERSITY

(Established under Galgotias University Uttar Pradesh Act No. 14 of 2011)

DEGREE

Session 2025-26
in

Computer Science and Engineering

By
Gauri Pachauri-22SCSE1011577
Harshit Rathore-22SCSE1011589

Under the guidance of
Dr.Firoj Ahmad

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GALGOTIAS UNIVERSITY, GREATER NOIDA

INDIA

Nov, 2025

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING

GALGOTIAS UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the project, entitled "UHID: Unified Health" in partial fulfillment of the requirements for the award of the B. Tech. (Computer Science and Engineering) submitted in the School of Computing Science and Engineering of Galgotias University, Greater Noida, is an original work carried out during the period of Aug, 2025 to Jun 2026, under the supervision of Prof.Firoj Ahmad, Department of Computer Science and Engineering, of School of Computing Science and Engineering , Galgotias University, Greater Noida.

The matter presented in the thesis/project/dissertation has not been submitted by me/us for the award of any other degree of this or any other places.

Student Names (Admission No.)

Gauri Pachauri-22SCSE1011577

Harshit Rathore-22scse1011589

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Guide Name

Designation

CERTIFICATE

This is to certify that Project Report entitled “UHID: Unified Health ID.” which is submitted by Gauri Pachauri and Harshit Rathore in partial fulfillment of the requirement for the award of degree B.Tech. in Department of School of Computing Science and Engineering , Galgotias University, Greater Noida, India is a record of the candidate own work carried out by him/them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree

Signature of Examiner(s) **Signature of Supervisor(s)**

External Examiner **Signature of Program Chair**

Date: 12 NOV 2025

Place: Greater Noida

ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B.Tech. Final Year. We owe special debt of gratitude to Professor Dr.Firoj Ahmad, Department of Computer Science & Engineering, Galgotias University, Greater Noida, India for his constant support and guidance throughout the course of our work. His/Her sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.

We also take the opportunity to acknowledge the contribution of Professor (Dr.) Harish Kumar GR, Dean, Department of Computer Science & Engineering, Galgotias University, Greater Noida, India for his full support and assistance during the development of the project.

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

Signature:

Gauri Pachauri-22SCSE1011577

Harshit Rathore-22SCSE1011589

Date : 12 NOV 2025

ABSTRACT

The Unified Health ID System (UHID) project aims to develop a secure, consent-based digital platform to unify patients' medical identities and health records under one unique identification number. The need for this project arises from the current challenges in India's healthcare system, where fragmented and unorganized patient data often results in delayed treatment, data loss, and duplication of medical histories. The system seeks to simplify healthcare access by introducing a digitally verified identity similar to academic or financial identifiers but designed exclusively for medical purposes. The project addresses this problem by designing a web-based platform using the MERN stack integrated with PostgreSQL and Prisma ORM, ensuring scalability, security, and privacy. Aadhaar-based duplicate prevention, role-based authentication, and encrypted data storage form the core of its design. Though the system's implementation phase is yet to begin, the architecture, database schema, and security framework have been successfully developed, meeting the design objectives and laying a strong foundation for future deployment within India's digital health ecosystem.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
Declaration	-	2
Certificate	-	3
Acknowledgement	-	4
Abstract	-	5
List of Figures	-	7
List of Abbreviations	-	8
CHAPTER 1	Introduction	10-12
	Overview and Background	
	Objectives of the Project	
CHAPTER 2	Software Requirement Specification (SRS)	13-21
	Functional and Non-Functional Requirements	
	System Interfaces and Data Model	
CHAPTER 3	System Design	23-28
	Architecture and Design Diagrams	
	Database Schema and Structure	
CHAPTER 4	Implementation and Results	29-31
	Software and Hardware Requirements	
	Implementation Details	
CHAPTER 5	Conclusion	32
	Performance Evaluation	
	Future Directions	
Appendix	Prisma Schema Structure	33-35
References	-	36

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 3.1	System Architecture of Unified Health ID System	24
Figure 3.2	UML Class Diagram of UHID System	25
Figure 3.3	Data Flow Diagram (DFD) of UHID System	26
Figure 3.4	Activity Diagram for Registration and Login Process	27
Figure 3.5	Database Schema Diagram using Prisma ORM	28

LIST OF ABBREVIATIONS

Abbreviation	Full Form
UHID	Unified Health ID System
ABDM	Ayushman Bharat Digital Mission
ABHA	Ayushman Bharat Health Account
NCD	Non-Communicable Diseases
ICMR	Indian Council of Medical Research
MoHFW	Ministry of Health and Family Welfare
NPCDCS / NP-NCD	National Programme for Prevention and Control of Non-Communicable Diseases
WHO	World Health Organization
AAM	Ayushman Arogya Mandir
Aadhaar	Unique Identification Authority of India (UIDAI) ID Number
EHR	Electronic Health Record
API	Application Programming Interface
ORM	Object-Relational Mapping
JWT	JSON Web Token
DB	Database
SQL	Structured Query Language
IDE	Integrated Development Environment
UI	User Interface
RBAC	Role-Based Access Control
TLS	Transport Layer Security

Abbreviation	Full Form
OTP	One-Time Password
UUID	Universally Unique Identifier
PBCR	Population-Based Cancer Registry
HBCR	Hospital-Based Cancer Registry

CHAPTER 1

INTRODUCTION

1.1 Background and Problem Context

Based on the comprehensive review of documents published by the Government of India through NITI Aayog, the World Health Organization (WHO), and the National Health Policy 2025, approximately 66% of deaths in India are attributed to Non-Communicable Diseases (NCDs). According to the Indian Council of Medical Research (ICMR) and the Ministry of Health and Family Welfare (MoHFW), India's disease burden has shifted significantly from communicable to chronic illnesses. Factors such as industrialization, urbanization, sedentary lifestyles, and an aging population have contributed to this shift.

The findings presented in this study are derived from three core national and international datasets:

1. **Cancer Data:**
   The 2025 ICMR analysis, based on 43 Population-Based Cancer Registries (PBCRs) and Hospital-Based Cancer Registries (HBCRs), revealed alarming growth in the incidence of oral, lung, and prostate cancers in men, and breast, cervical, and ovarian cancers in women.
   The Age-Adjusted Incidence Rate (AAIR) for males in Mizoram is 256.1 per 100,000, the highest recorded nationally, highlighting regional disparities.
2. **Mortality Data:**
   The 2025 *Lancet* global study analyzed data from 185 countries (2001–2019) and concluded that India's probability of premature death due to NCDs increased, while most other nations recorded declines. India's Mortality-to-Incidence Ratio (MIR) for cancer stood at 64.47%, nearly triple that of the United States (23.81%) and significantly higher than China (50.57%) and Indonesia (58.66%). This demonstrates severe systemic gaps in early diagnosis and treatment.
3. **Policy and Performance Data:**
   The MoHFW's NPCDCS/NP-NCD Progress Report (2024–2025) shows large-scale screening through Ayushman Arogya Mandirs (AAMs). By August 2025, over 38.96 crore individuals were screened for hypertension, 38.77 crore for diabetes, and 32.81 lakh for oral cancer. However, treatment linkage remains suboptimal, especially among low-income populations.

1.2 Key Observations

The ICMR report highlights a gender-specific cancer pattern:

- Males: Oral, lung, and prostate cancers dominate, with tobacco-related cancers contributing to 27.1% of the male cancer burden.
- Females: Breast, cervical, and ovarian cancers are most prevalent. Although cervical cancer rates are declining due to vaccination programs, breast cancer is increasing in nearly all registries.

The Longitudinal Ageing Study in India (LASI) reveals that NCD-related mortality is highest among poorer adults aged 45+, emphasizing that chronic diseases are no longer limited to affluent sections. The lack of early diagnosis, financial limitations, and poor treatment continuity contribute to this trend.

1.3 Digital Health Awareness Gap

Despite government efforts through the Ayushman Bharat Digital Mission (ABDM) and the ABHA ID initiative, digital health adoption remains low.
A recent survey conducted among 56 young adults (ages 16–25) revealed:

- 37.50% are fully aware of the ABHA ID.
- 46.40% have partial knowledge.
- 16.04% are entirely unaware.
  Only 15 respondents (≈25%) actually possess an ABHA ID.

This lack of awareness among youth — the most digitally literate demographic — indicates a critical communication gap. While the public supports the concept of digital health infrastructure, the government's execution and education strategy have not reached the grassroots level effectively.

1.4 Need for a Unified Health Identity System

The above data underscores the need for synchronized, organized, and continuously updated health data across India. Establishing a nationally accepted health identity, similar to APAAR ID for education, would enable lifelong tracking of health records from birth to old age.

Such a system would:

- Allow patients and doctors to access medical history instantly, reducing diagnostic delays.
- Replace paper-based records with a secure, digital, and interoperable system.

- Help the government and researchers analyze national health trends, identify high-risk areas, and predict emerging threats such as pandemics.
- Enable evidence-based policy formulation through accurate, real-time health statistics.

1.5 Proposed Solution: UHID – Unified Health ID System

The Unified Health ID System (UHID) is proposed as a technical solution to the systemic issues identified above. It aims to:

- Assign each patient a unique health identifier linked with Aadhaar to prevent duplication.
- Store health data digitally and enable retrieval across hospitals and states.
- Provide a role-based portal system for patients, doctors, and government administrators.
- Ensure data privacy, security, and consent-based sharing using advanced encryption and authentication mechanisms.

By creating an interoperable backend infrastructure using Node.js, Express.js, Prisma ORM, and PostgreSQL, the project lays the foundation for a scalable and secure national health data framework.

1.6 Summary

In conclusion, India's growing burden of chronic diseases, coupled with poor digital awareness and fragmented data systems, necessitates a centralized digital health identity platform. The UHID System addresses this challenge by bridging healthcare accessibility, improving data reliability, and supporting national health research and policymaking.

CHAPTER 2

SOFTWARE REQUIREMENT SPECIFICATION

2.1 Purpose

This SRS describes the software requirements for the UHID: Unified Health ID System. It documents the product perspective, functional and non-functional requirements, user characteristics, interfaces, constraints, assumptions, and key use cases. The SRS will be used by developers and stakeholders to guide design, implementation, testing, and deployment during Phase-1.

2.2 Product Perspective

UHID is a web-based identity and access management system for healthcare. It is designed as an independent software product with potential future integration into national health frameworks (e.g., ABHA/NDHM). The system uses a standard 3-tier architecture:

Presentation Layer: React.js web client for Patients, Doctors, and Admin.

Application Layer: REST API built with Node.js and Express; authentication and business logic reside here.

Database Layer: PostgreSQL accessed via Prisma ORM, storing normalized entities (Patient, Doctor, Admin, Consent, UHID records).

Phase-1 (this SRS) focuses on secure onboarding, unique UHID generation, authentication, basic consent management, and admin verification workflows.

2.3 System Interfaces

2.3.1 User Interfaces

Patient Portal (Web):

Registration form (personal details, Aadhaar reference, contact, DOB)

Login / Forgot Password

Dashboard showing UHID, consents granted, profile

Consent management screen (grant/revoke)

Doctor Portal (Web):

Registration form (personal details, license number, specialization)

Login / status (verified / pending)

Request access to patient data (triggers patient consent flow)

Admin Panel (Web):

Doctor verification dashboard (approve/reject documents)

User management (view logs, search users)

Audit trail and basic metrics

UI forms must validate input client-side (React) then server-side (backend).

2.3.2 Software Interfaces

REST API (Backend): JSON over HTTPS. Endpoints described in Section 2.7 (functional mapping).

Database: PostgreSQL v13+ accessed via Prisma ORM.

Third-party Services (optional / future):

Aadhaar verification API (government or licensed aggregator) — for duplicate prevention and verification.

Email / SMS gateway for OTP, notifications.

Logging / monitoring service (e.g., centralized ELK or cloud logging).

2.3.3 Hardware Interfaces

No special hardware. System runs on standard cloud/VM instances; clients are ordinary web browsers (desktop/mobile).

2.3.4 Communications Interfaces

HTTPS (TLS 1.2/1.3) for all client↔server and inter-service communication.

RESTful JSON payloads; standard HTTP status codes.

2.4 Functional Requirements

Each requirement is labeled (FR#) for traceability. These requirements are currently **in the design and development stage** and will be implemented in subsequent phases.

FR1 — Patient Registration

FR1.1: User can submit patient registration with fields: name, aadhaarNumber, dateOfBirth, email, contactNumber, address, password.

FR1.2: System must validate Aadhaar format server-side and check for duplicates (aadhaarNumber unique constraint).

FR1.3: On success, system generates a unique UHID and returns confirmation.

FR1.4: System stores password as salted hash (bcrypt) and never stores plaintext passwords.

FR2 — Doctor Registration

FR2.1: Doctor submits name, email, licenseNumber, specialization, password, and uploads verification documents.

FR2.2: System marks doctors as verified = false until admin approval.

FR2.3: Admin can approve/reject doctor registration; upon approval, doctor is notified.

FR3 — Authentication

FR3.1: Login uses email + password for all roles; upon authentication a JWT (access token) is issued with role claims and expiry.

FR3.2: Refresh token mechanism supported (optional) to allow session renewal.

FR3.3: Password reset via secure email OTP flow.

FR4 — UHID Generation & Management

FR4.1: Each successful patient registration triggers UHID generation with a stable, collision-resistant algorithm (e.g., UHID prefix + UUID or structured code).

FR4.2: UHID is stored in Patient table and is immutable.

FR5 — Duplicate Prevention

FR5.1: System enforces database unique constraints on aadhaarNumber and email.

FR5.2: Additional duplicate detection rule: near-duplicate detection on name + DOB + contact to flag probable duplicates for manual review.

FR6 — Consent Management

FR6.1: Patient can grant a doctor access to records for a specified period (grant request includes doctorId, patientId, expiry).

FR6.2: Consent entity records grantedAt, expiresAt, and scope (e.g., "basicProfile", "diagnostics", "prescriptions").

FR6.3: Doctor access is allowed only when an active consent exists.

FR7 — Admin Functions

FR7.1: Admin can view pending doctor verifications and accept/reject them.

FR7.2: Admin can search users, view audit logs, and flag suspicious behavior.

FR8 — Audit & Logging

FR8.1: Record all login attempts (success/fail), UHID creation events, consent grants/revocations, and doctor verification actions with timestamps and userId.

FR9 — API Validation & Error Handling

FR9.1: All API endpoints must validate input and return standard error codes and error messages in JSON.

2.5 Non-Functional Requirements (NFRs)

NFR1 — Security

All data in transit must use TLS.

Passwords hashed with bcrypt (cost factor >= 12).

JWT tokens signed with secure secret; access tokens short-lived (e.g., 15–60 minutes).

Sensitive fields (Aadhaar) may be stored encrypted or tokenized at rest (field-level encryption recommended).

Role-based access control (RBAC) enforced for endpoints.

NFR2 — Privacy & Compliance

Personal health data handling must follow applicable Indian laws and guidelines (data minimization, purpose limitation).

Patient consent must be auditable.

Data retention policies must be configurable.

NFR3 — Performance

Average registration API response time < 300ms under normal load.

System must support at least 500 RPS in phased horizontal scaling (cloud deployment).

NFR4 — Scalability

Application should be stateless (JWT sessions) to allow horizontal scaling.

Database designed with indexing on frequently queried fields (aadhaarNumber, email, createdAt).

NFR5 — Availability

Target 99.9% uptime for authentication services in production (with load balancing and health checks).

NFR6 — Maintainability

Code structured into modular controllers, services, and data access layers.

Unit and integration tests covering critical flows (registration, login, consent).

NFR7 — Auditability

All sensitive operations produce immutable logs with user, action, timestamp, and IP.

NFR8 — Usability

Simple registration forms with clear field validation messages; average registration time < 5 minutes for a typical patient.

NFR9 — Backup & Recovery

Daily database backups; point-in-time recovery for last 7 days. Backup testing performed at least monthly.

2.6 User Characteristics

Patients: Non-technical users; expect clear guided forms; may require local language support later.

Doctors: Technical/medical professionals; require quick verification and access to patient data upon consent.

Admin: Technical users (developer/IT staff); require dashboards and audit logs.

2.7 Constraints

Regulatory: Any Aadhaar verification must follow government rules; direct storage of Aadhaar may be restricted — use hashed/tokenized references unless allowed.

Connectivity: Real-time verification (Aadhaar or SMS OTP) requires stable network connectivity.

Third-party dependencies: SMS/email gateways, Aadhaar aggregator APIs may affect availability/latency.

Time & Scope: Current deliverable (Phase-1) excludes EHR storage, analytics, and mobile app.

2.8 Assumptions and Dependencies

A legal Aadhaar verification interface or acceptable alternative is available (or Aadhaar used only for duplicate prevention without storing raw Aadhaar number).

SMTP / SMS gateway credentials are available for OTP and notifications.

Prisma will be used as the ORM layer with PostgreSQL as RDBMS.

Frontend (React) will consume the documented REST APIs.

2.9 Apportioning of Requirements

Phase-1 (current): FR1–FR9 (Registration, Auth, UHID generation, Doctor verification, basic consent).
Deferred to Phase-2 / future: full EHR storage, advanced analytics, cross-facility record syncing, mobile app, integration with national registries.

2.10 Use Cases

Use Case 1: Register Patient

Use Case ID: UC-P01

Primary Actor: Patient

Preconditions: Internet access; patient has Aadhaar for verification.

Trigger: Patient opens registration page and submits form.

Basic Flow:

Patient fills registration form and submits.

Frontend validates fields and sends POST /api/patients/register.

Backend validates data, checks unique aadhaarNumber/email.

If unique, password is hashed and patient record created in DB.

System generates UHID and returns success with UHID (and email OTP for confirmation).

Alternate Flows:

A1: Duplicate Aadhaar/email → API returns 409 Conflict with message.

A2: Aadhaar validation fails → flagged for manual review.

Postconditions: Patient record created, UHID issued, audit log recorded.

Use Case 2: Register Doctor & Verification

Use Case ID: UC-D01

Primary Actor: Doctor; Secondary Actor: Admin

Preconditions: Doctor has valid license and documents.

Basic Flow:

Doctor fills registration form, uploads documents via POST /api/doctors/register.

System creates doctor record with verified = false.

Admin reviews documents in the Admin Panel and approves/rejects.

If approved, verified = true, notification sent to doctor.

Alternate Flows:

A1: Documents insufficient → Admin rejects with reason.

Postconditions: Verified doctor can request patient access and be visible as verified to patients.

Use Case 3: Login / Authentication

Use Case ID: UC-A01

Primary Actor: Any user (Patient, Doctor, Admin)

Preconditions: User registered and verified (for doctors, verified true).

Basic Flow:

User submits credentials to POST /api/auth/login.

Backend verifies password hash; if successful, issues JWT with role claims and expiry.

Frontend stores token in secure storage (HttpOnly cookie or secure storage) and uses it for subsequent calls.

Alternate Flows:

A1: Invalid credentials → 401 Unauthorized; audit logged.

A2: Account locked after multiple failed attempts (rate limiting) → 423 Locked.

Postconditions: Authenticated session established; audit log recorded.

Use Case 4: Grant Consent

Use Case ID: UC-C01

Primary Actor: Patient; Secondary Actor: Doctor

Preconditions: Doctor requests access or patient initiates consent grant.

Basic Flow:

Doctor requests access via POST /api/consents/request (or patient grants via UI).

Patient receives notification and approves via UI POST /api/consents/grant with doctorId and scope.

System creates Consent record with grantedAt, optional expiresAt.

Alternate Flows:

A1: Patient denies → Consent record not created; doctor notified.

A2: Patient revokes earlier consent → Consent marked revoked; audit logged.

Postconditions: If consent active, doctor can access specified data for duration.

2.11 API Endpoints (Phase-1 mapping)

The following endpoints are part of the planned API architecture for Phase-1 and are currently in the design stage (not yet implemented).

List of important backend endpoints (representative):

POST /api/patients/register — create patient (FR1)

POST /api/patients/confirm-otp — confirm registration OTP

POST /api/doctors/register — create doctor (unverified)

GET /api/admin/doctors/pending — admin: list pending verifications

POST /api/admin/doctors/:id/verify — admin verifies doctor (FR7)

POST /api/auth/login — authenticate and return JWT (FR3)

POST /api/auth/refresh — refresh token

POST /api/consents/request — doctor requests consent

POST /api/consents/grant — patient grants consent (FR6)

GET /api/patients/:uhid — get patient basic profile (doctor requires consent)

Each endpoint must enforce authorization middleware that validates JWT and role scopes.

2.12 Data Model (High level)

(Prisma models summarized)

Patient (id, name, aadhaarNumber, dateOfBirth, email, contactNumber, passwordHash, createdAt, UHID)

Doctor (id, name, email, licenseNumber, specialization, passwordHash, verified, createdAt)

Consent (id, patientId, doctorId, grantedAt, expiresAt, scope)

Admin (id, username, passwordHash)

AuditLog (id, userId, action, timestamp, meta)

Database indices: aadhaarNumber UNIQUE, email UNIQUE, index on createdAt.

2.13 Validation & Acceptance Criteria

Registration Acceptance:

Patient registration returns 201 Created and UHID on valid input.

Duplicate Aadhaar/email returns 409 Conflict.

Authentication Acceptance:

Valid credentials return 200 + JWT; invalid returns 401.

Doctor Verification Acceptance:

Admin approval sets verified=true and doctor can log in and access doctor flows.

Consent Acceptance:

Granting consent creates Consent row; doctor can access only permitted resources.

2.14 Traceability Matrix (brief)

Map of high-level requirements to modules:

FR1 → Patient Controller, Patient Service, Prisma Patient model

FR3 → Auth Controller, JWT Middleware, Token Service

FR6 → Consent Controller, Consent model, Authorization checks

FR7 → Admin Controller, Doctor model update

2.15 Future SRS Extensions (post Phase-1)

Planned additions for later phases:

EHR storage model with versioning and audit trail

Facility/hospital entity with multi-tenant record linking

Interoperability interfaces (HL7/FHIR)

Mobile app APIs and offline syncing

Data analytics and dashboards

CHAPTER 3

SYSTEM DESIGN

3.1 Overview

The System Design phase transforms the functional requirements defined in the SRS into a structured blueprint for implementation.
The UHID: Unified Health ID System follows a modular, scalable architecture that separates concerns between the frontend, backend, and database.

The architecture ensures:

Data security and privacy (via authentication, authorization, and encryption)

Ease of maintenance and scaling

Clear separation of responsibilities across modules and services

3.2 System Architecture

The system uses a Three-Tier Architecture approach:

1. **Presentation Layer (Frontend)**
    - Technology: React.js (Web App)
    - Role: Handles user interfaces for patients, doctors, and admin users.
    - Functions: Registration, login, dashboards, consent management.
    - Communication: Sends HTTP requests to the backend REST API.
2. **Application Layer (Backend)**
    - Technology: Node.js + Express.js
    - Role: Acts as the central controller for authentication, data validation, and logic.
    - Functions:
        - Manages patient/doctor registration and login.
        - Handles JWT-based authentication.
        - Connects with Prisma ORM for data persistence.
        - Manages admin verification and consent workflows (future phase).

3. **Database Layer**
    - Technology: PostgreSQL with Prisma ORM
    - Role: Stores structured, relational data for Patients, Doctors, Admins, and Consents.
    - Security: Enforces unique constraints (Aadhaar/email), hashed passwords, and relational integrity.

3.1 Architecture diagrams

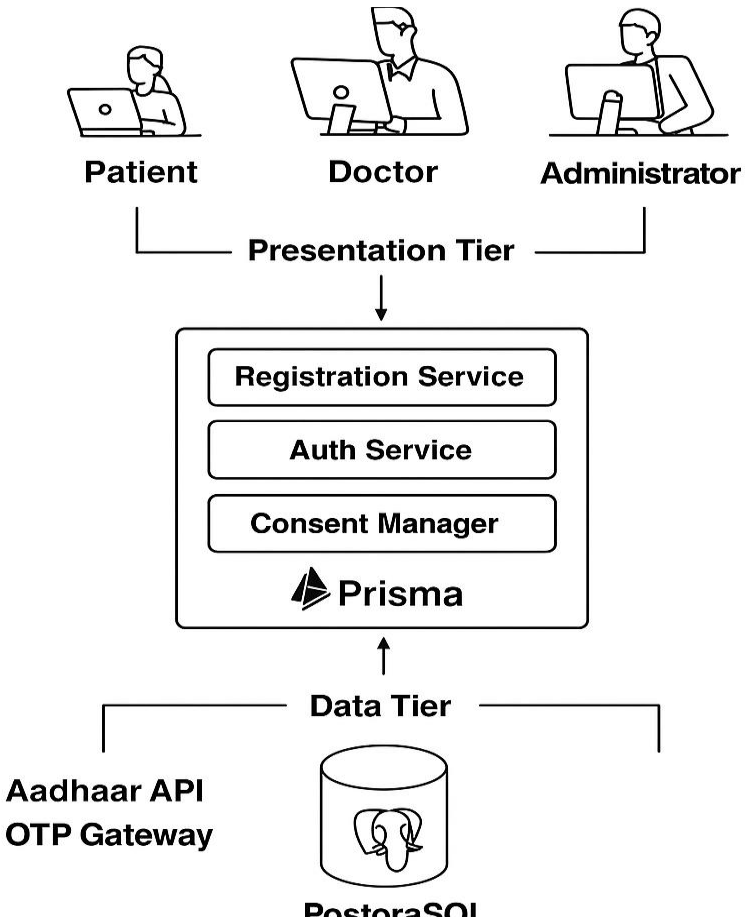
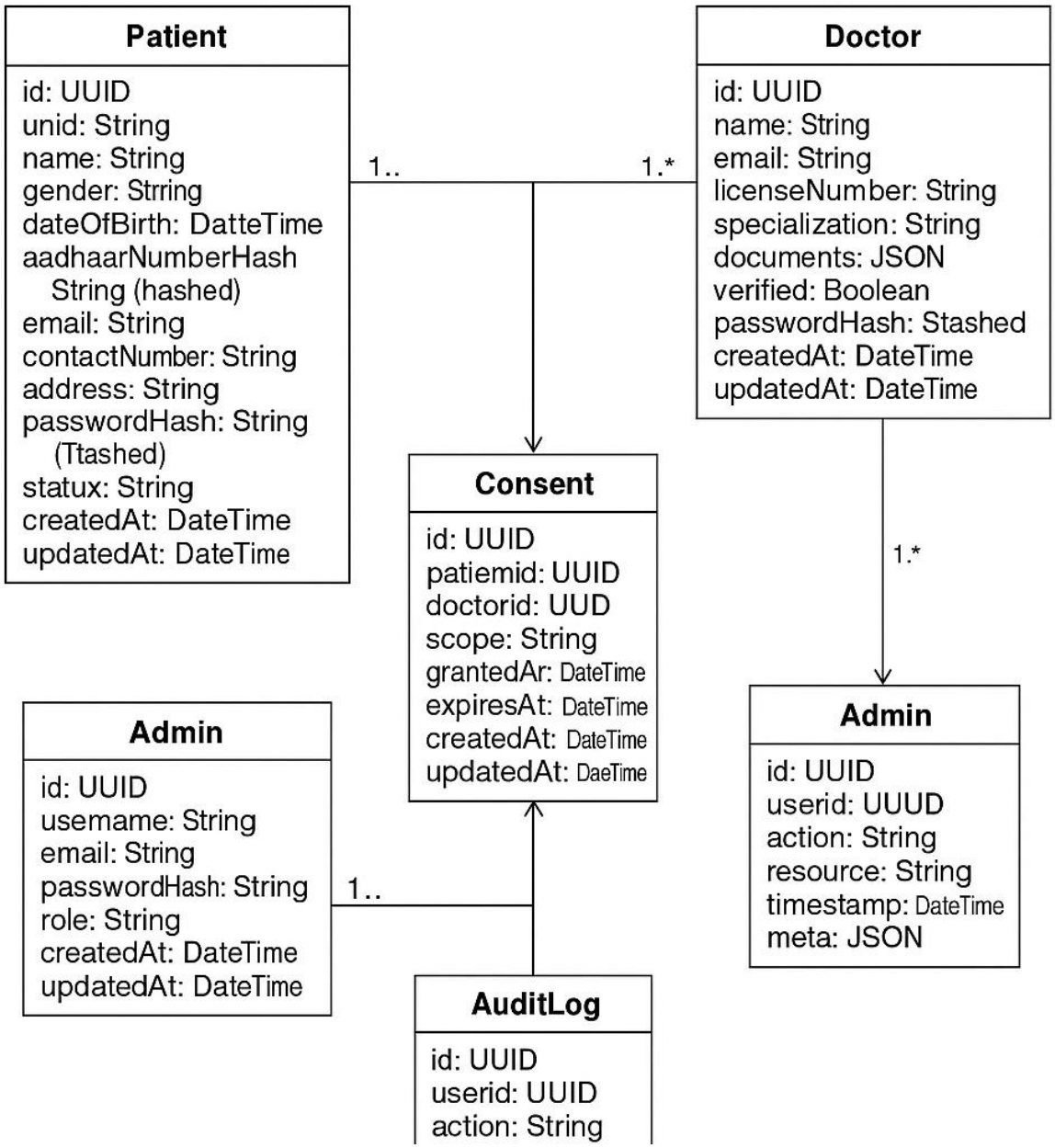
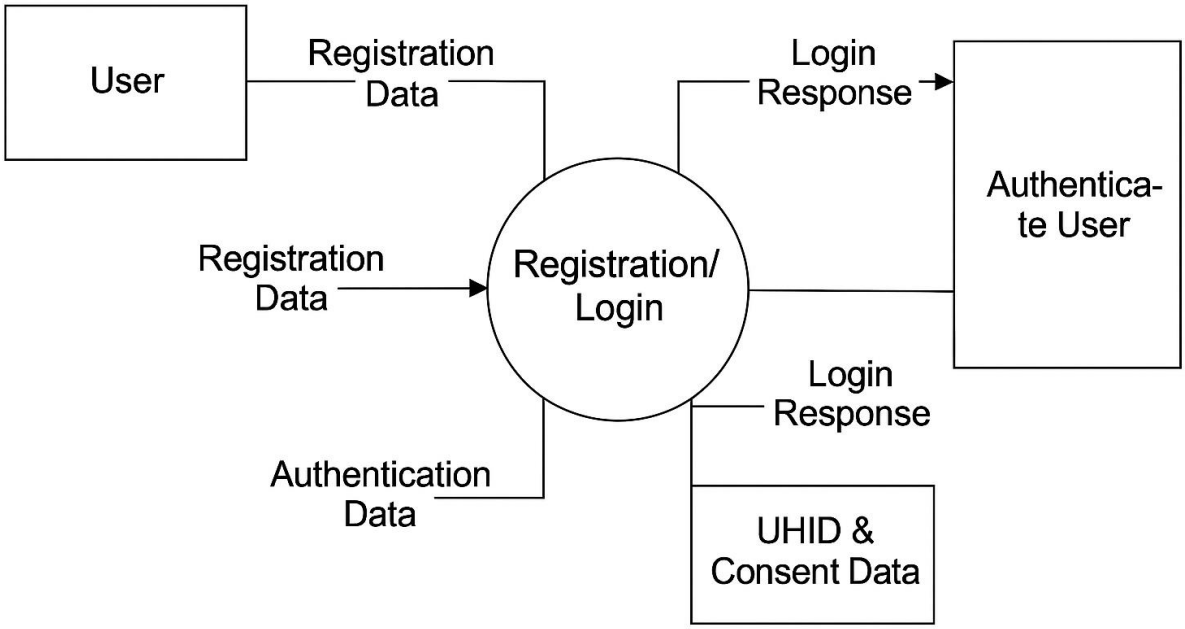


Figure 3.1 3-Tier Architecture Diagram example

3.2 Class diagrams

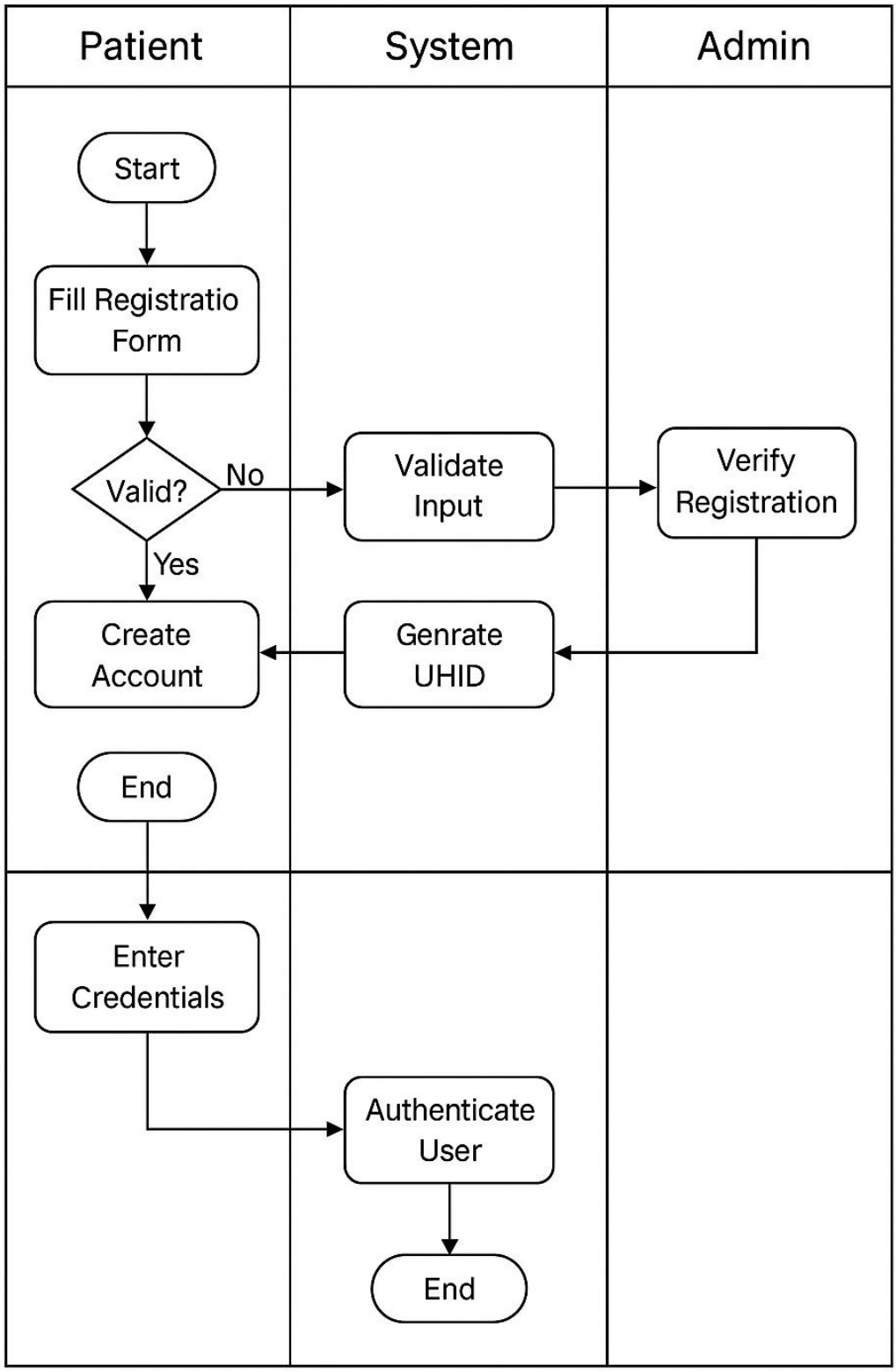


3.3 Data Flow Diagram

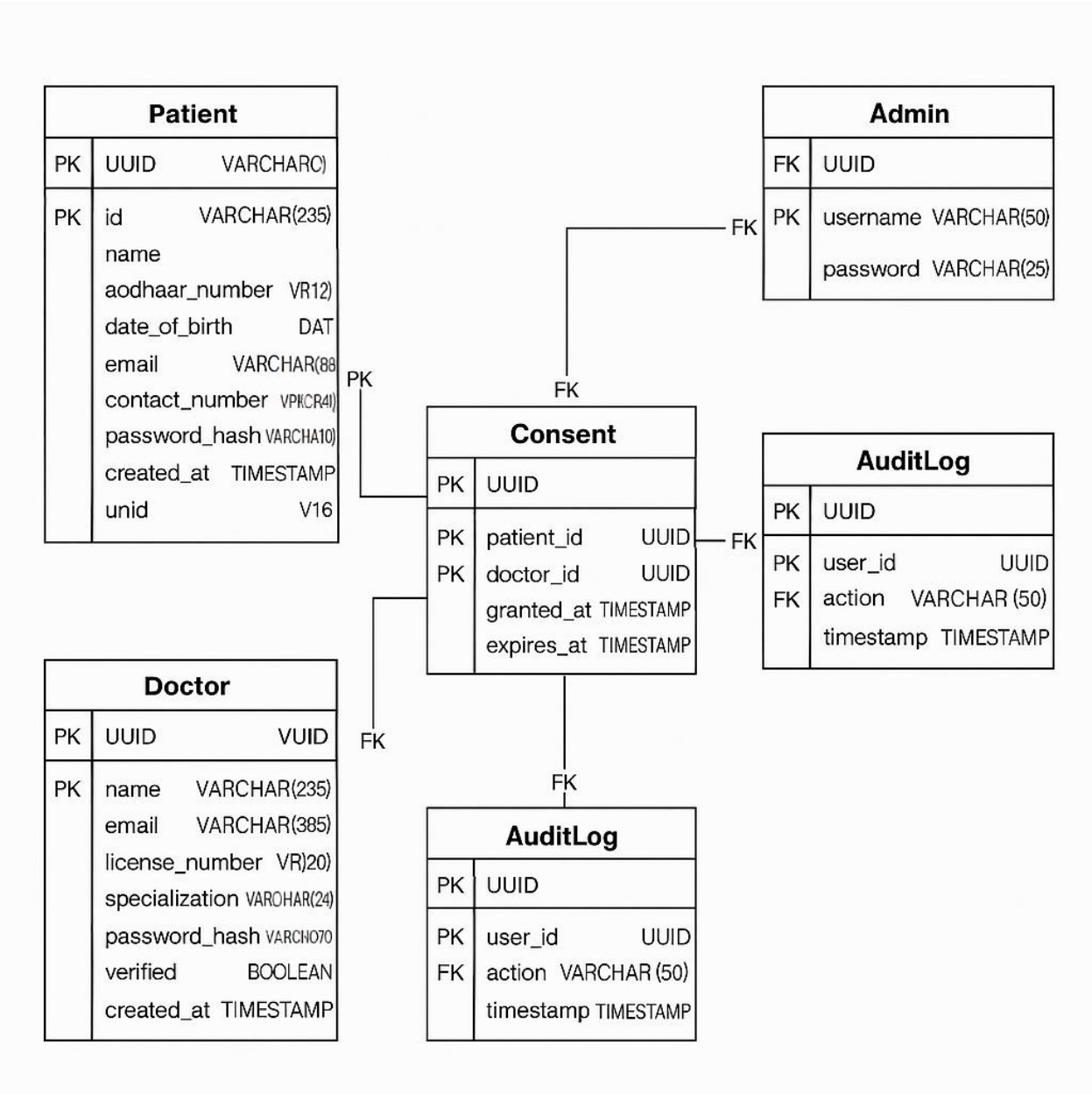


Data Flow Diagram
Level 1

3.4 Activity Diagram (Example for Registration and Login)



3.5 Database schema diagrams



CHAPTER 4

IMPLEMENTATION AND RESULTS

Software and Hardware Requirements

Software Requirements	Description
Operating System	Windows 10 / 11 (PowerShell environment)
Programming Language	JavaScript / TypeScript
Backend Framework	Node.js with Express.js
Frontend Framework	React.js (web-based interface)
Database	PostgreSQL (Version 13 or higher)
ORM Tool	Prisma ORM
Authentication	JSON Web Token (JWT)
Password Hashing	bcrypt
Development IDE	Visual Studio Code
Version Control	Git and GitHub
Testing Tools (Planned)	Postman, Jest
Browser Compatibility	Chrome, Edge, Firefox

Hardware Component	Minimum Requirement
Processor	Intel i3 / AMD equivalent or higher
RAM	8 GB (recommended for smooth development)
Storage	Minimum 50 GB free space
Display	1366 × 768 resolution or higher
Internet	Required for npm packages, Prisma, and DB setup

Assumptions and dependencies

Assumptions

- The system will be deployed on a secure cloud or server environment with stable internet connectivity.
- Each patient will have a valid Aadhaar number for duplicate prevention.
- Doctors will provide authentic license numbers and verification documents.
- Admins are verified technical users authorized to manage the system.
- All users will access the system through a modern web browser.

Dependencies

- Aadhaar verification API (Government or licensed aggregator) for patient verification.
- Email/SMS Gateway for OTP-based authentication.
- PostgreSQL Database Server for data storage and integrity.
- JWT Secret and .env Configuration for secure access management.
- Internet connectivity for API communication between frontend and backend.

Constraints

Constraint	Description
Regulatory	Must comply with Indian health data protection norms and Aadhaar data usage restrictions.
Data Privacy	Direct storage of Aadhaar numbers not allowed — only encrypted or hashed values.
Connectivity	Real-time Aadhaar or OTP verification requires stable internet.
Time Frame	Current work limited to Phase-1 (Authentication & Registration); full EHR integration will occur later.
External APIs	Delay or unavailability of Aadhaar/SMS API may affect verification workflow.

Results

As the project is currently in the **design and backend preparation** stage, system outputs and interface results are **not yet available**.
Results — including screenshots of successful registration, login, UHID creation, and consent management — will be incorporated after full implementation and testing of the system.

CHAPTER 5

CONCLUSION

5.1 Performance Evaluation

The **Unified Health ID System (UHID)** has been carefully designed with a modular three-tier architecture consisting of a React-based frontend, a Node.js and Express backend, and a PostgreSQL database managed through Prisma ORM. Although full implementation is yet to be completed, the planned structure ensures scalability, efficiency, and secure handling of data. Features such as JWT-based authentication, bcrypt password hashing, and Aadhaar-linked duplicate prevention are expected to provide strong performance in terms of speed, stability, and privacy once the system is deployed.

5.2 Comparison with Existing State-of-the-Art Technologies

When compared with national digital health initiatives like the Ayushman Bharat Health Account (ABHA) and the National Digital Health Mission (NDHM), the UHID system presents a simplified and developer-driven model. It focuses on user-controlled consent, secure authentication, and structured relational data management through Prisma and PostgreSQL. In contrast to large centralized systems, UHID emphasizes modularity and adaptability, making it easier to scale, maintain, and integrate with future digital-health platforms while ensuring transparency and patient privacy.

5.3 Future Directions

In future phases, the system will be extended to include complete electronic health-record (EHR) integration, hospital interoperability, and real-time analytics. The use of artificial intelligence for predictive health insights and blockchain-based data verification may further enhance reliability and trust. Practically, this work establishes the foundation for a unified, secure, and consent-driven digital health ecosystem that can reduce record duplication, streamline healthcare management, and support India's broader vision of a connected and transparent health infrastructure.

Appendix

Prisma Schema Structure

The following Prisma schema defines the database design for the Unified Health ID System (UHID).It represents the data models used to manage Patients, Doctors, Admins, Consents, and Audit Logs.The schema ensures relational integrity, privacy compliance, and scalability through PostgreSQL integration.

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

model Patient {
  id                String    @id @default(uuid())
  uhid              String    @unique
  name              String
  gender            String?
  dateOfBirth       DateTime?
  aadhaarHash       String
  email             String?   @unique
  contactNumber     String?   @unique
  address           String?
  passwordHash      String
  status            String    @default("PENDING")
  consents          Consent[]
  createdAt         DateTime  @default(now())
  updatedAt         DateTime  @updatedAt
}
```

```
model Doctor {
  id               String    @id @default(uuid())
  name             String
  email            String    @unique
  contactNumber    String?
  licenseNumber    String    @unique
  specialization   String?
  documents        Json?
  verified         Boolean   @default(false)
  passwordHash     String
  consents         Consent[]
  createdAt        DateTime  @default(now())
  updatedAt        DateTime  @updatedAt
}

model Admin {
  id            String   @id @default(uuid())
  username      String   @unique
  email         String?
  passwordHash  String
  role          String   @default("VERIFIER")
  createdAt     DateTime @default(now())
  updatedAt     DateTime @updatedAt
}
```

```
model Admin {
  id            String   @id @default(uuid())
  username      String   @unique
  email         String?
  passwordHash  String
  role          String   @default("VERIFIER")
  createdAt     DateTime @default(now())
  updatedAt     DateTime @updatedAt
}

model Consent {
  id         String   @id @default(uuid())
  patientId  String
  doctorId   String
  scope      Json?
  grantedAt  DateTime @default(now())
  expiresAt  DateTime?
  revokedAt  DateTime?
  createdAt  DateTime @default(now())
  updatedAt  DateTime @updatedAt

  Patient    Patient  @relation(fields: [patientId], references: [id])
  Doctor     Doctor   @relation(fields: [doctorId], references: [id])
```

```
  @@index([patientId])
  @@index([doctorId])
}

model AuditLog {
  id        String   @id @default(uuid())
  userId    String?
  action    String
  resource  String?
  ipAddress String?
  meta      Json?
  timestamp DateTime @default(now())
}
```

References

1. **NITI Aayog**, *National Health Policy 2025: Review of Non-Communicable Diseases in India*, Government of India, 2025.
2. **Indian Council of Medical Research (ICMR)**, *Population-Based Cancer Registries Annual Report*, 2025.
3. **World Health Organization (WHO)**, *Non-Communicable Diseases Country Profile: India*, 2025.
4. **Ministry of Health and Family Welfare (MoHFW)**, *National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD) Progress Report*, August 2025.
5. **Ayushman Bharat Digital Mission (ABDM)**, *ABHA ID Implementation Framework*, Government of India, 2024.
6. **Prisma ORM Documentation**, *Database Modeling for Node.js*, https://www.prisma.io/docs
7. **PostgreSQL Documentation**, *Data Modeling and Indexing*, https://www.postgresql.org/docs
8. **Express.js**, *Node.js Web Framework Guide*, https://expressjs.com
9. **JWT (JSON Web Token)**, *Introduction to JWT Authentication*, https://jwt.io/introduction
10. **WHO India Reports**, *National Burden of NCDs: Policy and Response*, 2024.

9/28/2025

Unified Health ID: Overcoming Fragmented Records & Low ABHA I'd Adoption

Harshit Rathore, Gauri Pachauri

ABSTRACT:

India is witnessing a steady rise in chronic illnesses such as diabetes, hypertension, heart disease, and cancer, which together account for a large share of nationwide deaths. Although initiatives like the Ayushman Bharat Digital Mission (ABDM) and ABHA health ID have been launched to strengthen the healthcare system, several barriers limit their effectiveness. Key issues include fragmented health records, low digital adoption in rural regions, and the lack of integrated approaches that support long-term recovery. These shortcomings create multiple challenges. Patients are often subjected to repeat diagnostic tests, delayed care, higher treatment costs, and limited participation in managing their own health. Doctors lack continuous access to patient histories, while researchers struggle to obtain structured and anonymized data for meaningful studies. To overcome these gaps, this project proposes a patient-centric digital health identification system. The system would consolidate lifelong medical records from birth, support secure and consent-based data sharing, and include personalized, evidence-driven lifestyle and natural care suggestions in addition to conventional treatment. A web-based patient portal and clinician dashboard will be developed to demonstrate functionality and usability. Anticipated outcomes include improved continuity of care, reduced duplication in testing and spending, empowered patients through personalized insights, and the availability of reliable datasets for medical research and policymaking. By bridging these gaps, the project aims to strengthen India's digital health ecosystem and support sustainable development goals.

INTRODUCTION:

Based on the comprehensive review of documents published by the government of India though NITI Ayog, WHO and the National Health Policy 2025 approximately 66% of death are due to NCD (Non- Communicable Diseases). As per the Indian Council of Medical Research (ICMR), and the Ministry of Health and Family Welfare, India's health burden is already shifted from communicable diseases to the chronic diseases and the reason being industrialisation, urbanization, lifestyle changes, and an aging population.

The findings presented herein are synthesized from three core national and international data streams:

1. **Cancer Data:** Incidence and prevalence metrics are derived from the 2025 ICMR analysis, which incorporated information from 43 Population-Based Cancer Registries (PBCRs) and Hospital-Based Cancer Registries (HBCRs). This analysis provides Age-Adjusted Incidence Rates (AAIR), lifetime risk assessments, and details on site-specific malignancy prevalence.
2. **Mortality Data:** The comparative assessment of mortality trends relies on the 2025 *Lancet* study. This global analysis reviewed mortality data from 185 countries from the year 2001 to 2019, quantifying the change in the probability of death from NCDs between birth and age 80 years in India specifically.
3. **Policy and Performance Data:** The evaluation of government response utilizes the NPCDCS/NP-NCD progress reports released by the MoHFW (Ministry of Health and Family Welfare) as of August 2025. This data details the cumulative screening throughput, diagnosis rates, and treatment linkage for five major NCDs (Hypertension, Diabetes, Oral, Breast, and Cervical Cancers) among the target population aged 30 years and above, largely conducted at Ayushman Arogya Mandirs (AAMs).

The ICMR analysis also identified the clear pattern gender-specific patterns in cancer incidence:

- **Males:** The three most common cancers identified nationwide are **oral, lung, and prostate** cancers. The risk associated with oral cancer is particularly worrying, showing significant increases in incidence rates across 14 population-based cancer registries (PBCRs). Tobacco-related cancers remain the single largest category contributing to the male cancer

burden, estimated to contribute 27.1% of the total cancer burden based on 2020 projections. Cancers of the gastrointestinal tract also contribute significantly.

- **Females:** The leading sites of malignancy are **breast, cervical, and ovarian** cancers. While cervical cancer shows a declining trend in most registries, potentially indicating the success of vaccination and screening efforts, breast cancer incidence rates have increased significantly across many registries.

Key Epidemiological Metrics of Cancer Burden in India (2025 ICMR Analysis)

Metric	**National Average/Key Finding**	**Extreme Regional Outlier**	**Specific AAIR (per 100,000)**
Lifetime Risk (All Cancer)	11.0%	Mizoram (Males)	21.1%
Highest Age-Adjusted Incidence Rate (AAIR)	N/A	Aizawl District (Males)	256.1(95% CI, 245.2–267.0)
Most Common Cancers (Males)	Oral, Lung, Prostate	N/A	Oral cancer increasing significantly
Most Common Cancers (Females)	Breast, Cervical, Ovarian	N/A	N/A

Comparative Analysis of India's Mortality Ratio

The most definitive evidence of systemic failure in India's cancer control framework is its excessively high cancer Mortality-to-Incidence Ratio (MIR). While high incidence rates drive public awareness, the MIR serves as the critical measure of the public health system's ability to provide timely diagnosis and effective treatment, thereby saving lives.

Quantifying Poor Outcomes: India's MIR in 2022 was **64.47%**, establishing it as the **highest ratio among the top 10 countries globally** ranked by absolute cancer incidence. This figure indicates that nearly two-thirds of all newly diagnosed cancer patients in India succumb to the disease.

This outcome stands in sharp contrast to the effectiveness of healthcare systems in high-income nations. For example, the United States, despite recording a greater absolute number of cancer incidences, maintains an MIR of only 23.81%. India's ratio is also higher than other major emerging economies, including China (50.57%) and Indonesia (58.66%).

Quantifying India's Outlier Status

The 2025 study in *The Lancet* delivered a sobering confirmation that India represents a dangerous outlier in the global trajectory of NCD control. Globally, between 2010 and 2019, four out of every five countries experienced a decline in the probability of premature death from chronic diseases. Conversely, this probability **increased in India** during the same decade.

In India, data from the Longitudinal Ageing Study in India (LASI) highlights a paradox regarding the economic distribution of NCD-related mortality. The percentage share of NCD-related mortality among adults aged 45+ is found to be greatest among the **poorer** income group (43.08%), followed by the richest group (39.54%). This distribution demonstrates that NCDs are no longer merely "diseases of affluence." While high-income groups may exhibit lifestyle risk factors, the poorer groups face mortality risks exacerbated by delayed diagnosis, lack of sustained treatment, and poor financial capacity for chronic disease management.

Policy Implementation and Program Performance: The NPCDCS Review (2024-2025)

NPCDCS Progress on Screening and Treatment (Target Population: Age:30+), As of August 2025

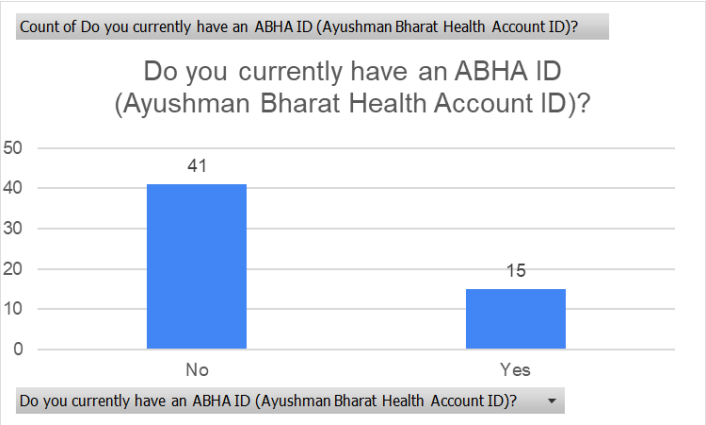
Indicator	**Hypertension (in crores)**	**Diabetes (in crores)**	**Oral Cancer (in lakhs)**	**Breast Cancer (in thousands)**	**Cervical Cancer (in thousands)**
Screening Conducted	38.96	38.77	32.81	17.21	10.21
Diagnosed Cases	5.13	3.46	1.88	41.39	90.29
Cases Under Treatment	4.75	2.84	1.77	61.74	97.09

While the Ayushman Bharat Digital Mission (ABDM) and the associated ABHA ID represent a vital governmental step towards establishing a unified, paperless healthcare ecosystem, their effectiveness hinges on public adoption and awareness. Recent survey data collected from 56 young adults (aged 16 to 25) offers insight into the nascent understanding of this critical digital infrastructure among a key demographic.

Awareness Gap The survey results reveal a significant gap in digital health literacy among young adults. Only 37.50% of the respondents reported being fully aware of the purpose and function of the ABHA ID. The largest group, 46.40%, possessed only partial knowledge, while 16.04% were entirely unaware.

This gap suggests that despite major national efforts, the communication strategy regarding the mechanics and utility of digital health accounts is failing to penetrate effectively among youth, who are often considered early adopters of technology.

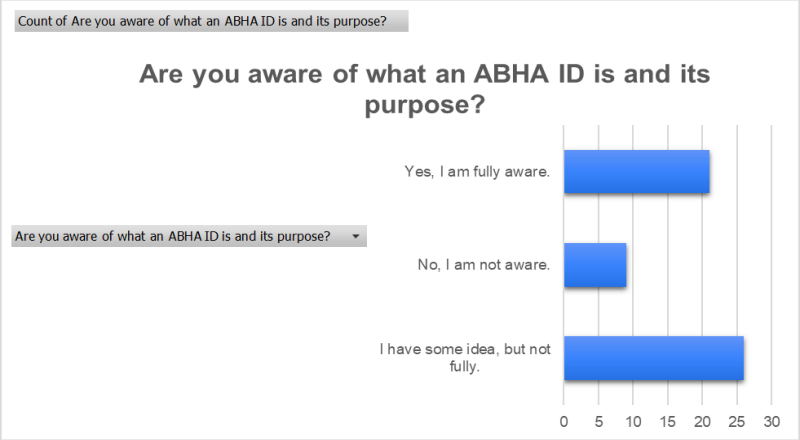
The survey shows that the awareness amongst the youth of India regarding this is very less.



Only 15 out of 56 have an ABHA I'd that is about 25% , it is depressing to see least awareness amongst the youth.

To understand whether the youth have a little knowledge about this cause and the need of such an initiative another survey question was, it's stats are as follows:

Are you aware of what an ABHA ID is and its purpose?	Numbers	Percentage
I have some idea, but not fully.	26	46.40
No, I am not aware.	9	16.04
Yes, I am fully aware.	21	37.50



This dichotomy presents a clear policy opportunity: the underlying faith in the digital health mission is robust, indicating public readiness for the concept. However, this high optimism risks translating into systemic disappointment if the government fails to convert potential into reality by closing the knowledge gap and ensuring the seamless, reliable integration of the ABHA ID into routine health services. The key challenge remains not the concept itself, but the tactical execution of awareness and the guarantee that digital accounts translate into tangible benefits, such as quicker access to records and reduced administrative friction.

Taking into consideration the above statics from the national , international organisations and the recent survey conduct during the research shows that there stands a need for synchronised, consecutive and organised data to prevent, regulate and monitor the heath trades of India which enables a whole some way to study, store and use the heath records of the patients.

The medical history if gets stored at an early age (i.e.- at the time of birth) would enable the patients to retrieve and organise the data in a way that can easily be accessed by them and the doctors as and when needed. This way will not only help reduce patients' time to find and present the medical reports but will also help to replace the paperwork to a much fast and easily accessible way. Keeping in mind the need to study the health of our country for further studies and research this data will also help to build a strong base and foundation for the same. The need for such a system arises when the doctor wants patients' medical history at the time of chronic diseases and usually the patients specially the ones' coming from a primitive area tend to provide incomplete medical history that hampers the recovery of the patients and lead to medication complexations. These aspects strengthen the concept of Nation wide excepted health I'd similar to APAAR I'd (I'd to store the academic history) should be made compulsory and that's what the project is about it helps to establish a portal for the doctors' , the government and the patients' . The project not only

focus on the patients' medical history to be stored digitally but also can easily be retrieved as and when need. Such an initiative not allows to reduce the patients' work to keep the medical reports save but also help the government study the daily trends of the country in the medical field which not only helps to create the policies that would benefit the people but also help the researches to understand which are the areas of major concern and this would help to predict the underlining threat of a pandemic that took place in the past.